

Deterministic Intent Gating, Sovereign Emergency Calling, and Two-Tier Civic Triage: An Empirical Evaluation of Non-Autoregressive Decision Models versus Foundation Large Language Models in Municipal 911/112 Operations and High-Throughput Public Administration

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Abstract—The wholesale integration of autoregressive Large Language Models (LLMs) into municipal emergency hotlines (e.g., 911 / Indonesia 112), public administration dispatch channels, and enterprise customer support has introduced severe operational vulnerabilities: queue divergence, catastrophic service-level agreement (SLA) breaches, non-deterministic schema mutation, and privacy exposure of personally identifiable information (PII). In life-safety emergency dispatch, sequential next-token synthesis incurs multi-second latency ($>4,500$ ms to $>12,500$ ms), severely compromising the 3-to-5 minute clinical “golden period” of out-of-hospital cardiac arrest and violating statutory dispatch timing regulations (<3 s). In this paper, we present an empirical evaluation conducted on **DecisionModelBench**, benchmarking non-autoregressive decision models against flagship foundation generative LLMs—including the Indonesian sovereign model Sahabat-AI 8B Instruct, Alibaba Cloud’s Qwen 2.5 7B Instruct, and Google DeepMind’s Gemma 2 9B Instruct deployed on dual NVIDIA Tesla T4 GPUs—across fifteen comprehensive scenarios: ten high-throughput civic and enterprise support cases and five critical life-safety 911/112 emergency calling scenarios. We formalize Erlang-C queue explosion dynamics, statutory SLA breach probability integrals, clinical survival probability decay functions, and schema entropy failure laws. Across civic triage, non-autoregressive decision models (TypeSafe JEV System One at 197.6 ms, OpenJev 0.5B at 540.6 ms, Kev-0.8B at 648.5 ms, and dedicated Laya 421M at 35.8–63.2 ms) achieve 100% schema compliance, up to 100% routing accuracy, zero token waste, and 100% SLA compliance (<500 ms) at \$0.05 per 100,000 queries. In critical emergency calling, Standalone LLMs exhibit dangerous dispatch latencies: 3,340.0 ms for Sahabat-AI 8B, 10,024.4 ms for Qwen 2.5 7B, and 12,681.7 ms for Gemma 2 9B, with Qwen exhibiting severe syntax fragility (33.3% standalone accuracy due to markdown fence wrapping). Conversely, our **Two-Tier Hybrid Tandem Pipeline** decouples physical CAD mobilization (<220 ms, up to $64.2\times$ acceleration, 100% deterministic accuracy, 0 tokens) from asynchronous high-fidelity conversational guidance (207–294 tokens), establishing the definitive paradigm for life-safety emergency operations. Furthermore, we demonstrate that non-autoregressive decision models filter out hoax and prank calls during crisis call storms in 223.5 ms without consuming GPU token generation, protecting municipal dispatch queues from catastrophic saturation.

Index Terms—Public Administration AI, Sovereign AI, Emergency Calling 911/112, Two-Tier Hybrid Brain Pipeline, Intent Gating, Non-Autoregressive Decision Models, Large Language Models, Erlang-C Queueing, Golden Period Resuscitation, Data Sovereignty, Total Cost of Ownership.



1 INTRODUCTION

MUNICIPAL governments, emergency response agencies, public utilities, and enterprise customer service centers are increasingly adopting artificial intelligence to process escalating volumes of citizen distress calls, incident reports, and support inquiries [1], [2]. National digital public infrastructure initiatives—such as Indonesia’s unified municipal emergency hotline 112, the national grievance

portal SP4N-LAPOR!, and regional disaster management networks (BPBD/Basarnas)—confront intense traffic surges during meteorological disasters, civil security incidents, and acute public health emergencies. In municipal emergency call centers, time is the single most critical determinant of clinical survival and property salvage: for an out-of-hospital cardiac arrest or acute respiratory failure, each thirty-second delay in dispatching advanced life support reduces resuscitation odds exponentially [15], [16].

To automate call intake, transcript classification, and departmental dispatch, public agencies and enterprise IT departments have predominantly turned to autoregressive

foundation Large Language Models (LLMs) [3]–[5]. Under this standard paradigm, inbound caller audio is converted to text and injected into complex prompt templates, prompting a 7B–70B parameter generative decoder to synthesize structured JSON dispatch payloads token by token:

$$\Pr(y_{1:L} \mid X) = \prod_{k=1}^L \Pr(y_k \mid y_{<k}, X) \quad (1)$$

While autoregressive LLMs possess rich linguistic fluency, deploying them as front-line operational dispatchers and real-time intake gates introduces severe architectural pathologies:

- 1) **Dangerous Dispatch Latency and Queue Explosion:** Synthesizing structured JSON requires $L \in [150, 300]$ sequential autoregressive forward passes across memory-bandwidth-constrained GPU accelerators. On enterprise hardware, this incurs an inference delay of 3,340 ms to 12,680 ms per call. In municipal emergency operations, statutory regulations (such as NFPA 1221, EENA, and Indonesian Ministry of Communication and Informatics standards for 112 hotlines) mandate initial dispatch within 3 seconds ($\tau_{\text{dispatch}} < 3.0$ s). Autoregressive LLMs inherently breach this threshold. Furthermore, under Erlang-C queue dynamics, multi-second service times trigger queue explosions during catastrophic surges (e.g., severe earthquakes or flash floods), exhausting server connection pools and leaving citizens stranded on hold.
- 2) **Schema Entropy and Syntax Brittleness:** Autoregressive token sampling is non-deterministic. Across long generation horizons, cumulative drift probabilities result in missing structural commas, unescaped quotation marks, or markdown fence wrapping, causing downstream Computer-Aided Dispatch (CAD) systems to throw parsing exceptions.
- 3) **Prohibitive Operational Cost and Energy Burn:** Generating 200 tokens per inquiry across hundreds of thousands of daily civic interactions imposes exorbitant compute expenditures and massive carbon footprints, rendering nationwide public deployment fiscally unsustainable.
- 4) **Data Sovereignty and PII Exposure:** Ingesting sensitive citizen records—such as National Identity Numbers (NIK), bank accounts, residential addresses, and medical conditions—into open generative attention contexts creates severe data leakage vectors, violating sovereign data protection mandates such as Indonesian Law No. 27 of 2022 on Personal Data Protection (UU PDP). Furthermore, foundation models remain vulnerable to prompt-injection jailbreaks that can force the exfiltration of internal system prompts and sensitive data.

To overcome these structural limitations, this paper investigates the application of **non-autoregressive decision models** as deterministic, sub-200ms semantic reflex gates (System 1) for public administration, municipal emergency dispatch, and enterprise support. Unlike generative decoders, non-autoregressive decision models project input

representations directly onto categorical action spaces and continuous urgency distributions in a single feed-forward pass ($\mathcal{O}(1)$), emitting typed mathematical tensors with zero token overhead.

Furthermore, recognizing that emergency callers require both instantaneous physical rescue dispatch and calming, empathetic verbal guidance, we evaluate a novel **Two-Tier Hybrid Tandem Pipeline** implemented on a distributed multi-GPU architecture (dual NVIDIA Tesla T4 GPUs). In this tandem configuration, Tier 1 (Decision Model) executes sub-200ms deterministic triage, dispatching rescue fleets immediately without generating tokens, while Tier 2 (Sahabat-AI 8B, Qwen 2.5 7B, or Gemma 2 9B) concurrently assimilates the structured triage metadata to synthesize rich, empathetic first-aid protocols (e.g., cardiopulmonary resuscitation rhythms, structural fire smoke evasion steps) asynchronously.

Using **DecisionModelBench**, we conduct an empirical evaluation comparing non-autoregressive models (TypeSafe JEV System One, OpenJev 0.5B, Kev-0.8B, and Laya Multilingual 421M) against prominent foundation generative models (Sahabat-AI 8B Instruct, Qwen 2.5 7B Instruct, and Gemma 2 9B Instruct). The primary contributions of this work are as follows:

- 1) **Rigorous Queue-Theoretic, SLA, and Resuscitation Formulations:** We formulate the Erlang-C queue waiting time and show why a transition from 197.6 ms to 5,107.8 ms service time causes exponential queue explosion under emergency loads. We formalize the statutory SLA breach probability integral, the schema entropy failure equation, and the clinical survival probability decay function governing out-of-hospital cardiac arrest golden periods.
- 2) **Comprehensive Fifteen-Scenario Civic and Emergency Benchmark:** We evaluate 10 realistic public administration and enterprise scenarios alongside 5 critical life-safety 911/112 emergency calling scenarios.
- 3) **Cross-Architecture Multi-LLM Tandem Evaluation on Sovereign Hardware:** We empirically validate the Two-Tier Hybrid Tandem Brain across three flagship open-weight LLMs (Sahabat-AI 8B, Qwen 2.5 7B, and Gemma 2 9B) on dual NVIDIA Tesla T4 GPUs. Tier 1 dispatches rescue fleets in 178.5 ms to 217.7 ms (< 220 ms, up to $64.2\times$ acceleration, 0 tokens), while Tier 2 synthesizes comprehensive conversational guidance in 207–294 tokens, establishing Pareto optimality across latency, safety, and citizen reassurance.
- 4) **Disaster Call Storm Prank Load-Shedding:** We demonstrate that non-autoregressive decision models filter out hoax and non-emergency calls in 223.5 ms without consuming GPU generation tokens, preventing queue saturation during mass-casualty crisis events.
- 5) **Empirical Performance, Cost, and Privacy Governance:** We demonstrate that non-autoregressive models achieve 100% schema compliance, up to 100% routing accuracy, and sub-200ms latency at \$0.05 per 100,000 queries. We expose the structural

syntax fragility of standalone LLMs (e.g., Qwen 2.5 suffering 33.3% accuracy due to markdown fence parsing crashes) and prove that non-autoregressive models act as deterministic semantic firewalls, eliminating PII exfiltration risks under Indonesian UU PDP No. 27/2022 while cutting operational inference costs by 99.87%.

2 RELATED WORK

2.1 Autoregressive Language Models in Automated Administration

Large Language Models built upon decoder-only Transformer topologies [6] have been widely proposed for automated document processing, conversational agents, and intent classification [1], [2]. To adapt these models for national administrative environments, sovereign language initiatives—such as Sahabat-AI in Indonesia [3] and SEA-LION in Southeast Asia—have pre-trained and fine-tuned 7B–8B parameter architectures on regional linguistic and cultural corpora.

However, executing decoder-only LLMs in production requires memory-bandwidth-bound autoregressive decoding [7]. Each emitted token requires transferring the model’s entire weight matrix Θ from high-bandwidth device memory (HBM/VRAM) to compute cores. For an 8-billion parameter model quantized to 4-bit precision ($|\Theta| \approx 4.5$ GB), generating a 200-token JSON dispatch payload demands reading approximately 900 GB of data across the memory bus. On enterprise accelerators such as the NVIDIA Tesla T4 (300 GB/s theoretical bandwidth), physical memory throughput strictly limits generation velocity to 25–35 tokens per second, guaranteeing a minimum latency of 4 to 6 seconds per request. While constrained decoding frameworks (such as Outlines [8] and Guidance) enforce regular-expression and context-free grammar constraints over the LLM vocabulary, they do not resolve the fundamental memory-bandwidth serialization bottleneck.

2.2 Non-Autoregressive Decision Architectures

Non-autoregressive neural networks discard sequential generation, computing representations simultaneously across all output positions [9]. Compact bidirectional encoder models, such as ModernBERT [10] and DeBERTa [11], process full input sequences in parallel, producing dense pooled embeddings $\mathbf{h} \in \mathbb{R}^d$ that can be mapped directly into discrete classification logits or continuous regression targets via lightweight linear projection heads:

$$\mathbf{z} = \mathbf{W}_h \mathbf{h} + \mathbf{b}_h \quad (2)$$

A complementary approach, instantiated in TypeSafe JEV System One [12], evaluates continuation logits over pre-specified candidate tokens or structured decision keys in a single forward pass without autoregressive loop execution. By evaluating:

$$\Pr(a_k | X) = \frac{\exp(z_{a_k})}{\sum_{j=1}^K \exp(z_{a_j})} \quad (3)$$

these models yield calibrated decision distributions within 50 ms to 200 ms. Because the output is mapped directly to statically typed schemas in host memory, non-autoregressive decision models guarantee 100% syntax validity and consume zero generation tokens.

2.3 Queueing Theory and Emergency Dispatch Standards

The application of queueing theory to civic emergency call centers and public administration dispatch is well established in operations research [13], [14]. In municipal emergency networks (e.g., 911 or 112 services), delayed dispatch directly correlates with increased property damage and mortality [15]. International and national emergency dispatch standards—such as NFPA 1221 (National Fire Protection Association standard for public safety communications), EENA (European Emergency Number Association), and the Indonesian Ministry of Communication and Informatics (Kominfo) regulations for emergency number 112—stipulate that alarm processing and dispatch determination must occur within strict operational bounds, typically requiring dispatch initiation within 3 to 15 seconds. When automated AI agents replace human call-takers, the stochastic service distribution of the AI system governs aggregate queue waiting times. Traditional queueing literature demonstrates that server utilization levels exceeding 85% in multi-server Markovian systems induce non-linear queue explosions. Introducing multi-second service times into high-volume intake pipelines violates the fundamental capacity constraints required to maintain stable civic dispatch operations.

3 MATHEMATICAL AND QUEUE-THEORETIC FORMULATION

3.1 Erlang-C Delay and Queue Explosion Dynamics

Consider a municipal public service intake gateway modeled as an $M/M/c$ queueing system, where inbound citizen reports arrive according to a Poisson process with aggregate arrival rate λ (queries per second). The processing infrastructure comprises c parallel compute workers (GPU/CPU execution instances). Each worker evaluates incoming reports with an independent and identically distributed exponential service duration with mean $\bar{t} = 1/\mu$, where μ denotes the service rate per worker.

The offered traffic load to the system, measured in Erlangs, is defined as:

$$a = \frac{\lambda}{\mu} = \lambda \cdot \bar{t} \quad (4)$$

The server utilization factor ρ across the c parallel workers is:

$$\rho = \frac{a}{c} = \frac{\lambda}{c\mu} \quad (5)$$

For system stability, the ergodicity condition requires that the incoming arrival rate does not exceed the aggregate processing capacity:

$$\rho < 1 \iff \lambda < c\mu \quad (6)$$

Under stable conditions ($\rho < 1$), the probability that an arriving citizen report finds all c workers busy and is forced to wait in the incoming queue is given by the Erlang-C formula:

$$C(c, a) = \frac{\frac{a^c}{c!} \frac{1}{1-\rho}}{\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!} \frac{1}{1-\rho}} \quad (7)$$

The expected queue waiting time W_q experienced by an inbound citizen complaint prior to AI evaluation is:

$$\begin{aligned} W_q &= \frac{C(c, a)}{c\mu(1-\rho)} \\ &= \frac{C(c, a)}{\lambda \left(\frac{1}{\rho} - 1\right)} = \frac{C(c, a)}{\lambda(1-\rho)} \end{aligned} \quad (8)$$

The total system response time T_{sys} , encompassing queue waiting delay and AI inference duration, is:

$$\begin{aligned} T_{\text{sys}} &= W_q + \bar{t} \\ &= \frac{C(c, a)}{c\mu(1-\rho)} + \frac{1}{\mu} \end{aligned} \quad (9)$$

Catastrophic Queue Explosion under Emergency Surge: Suppose a severe metropolitan flood or widespread emergency generates an arrival rate of $\lambda = 50$ requests/s. Consider an enterprise compute cluster provisioned with $c = 16$ parallel GPU workers:

- **Case 1: Non-Autoregressive Decision Model (JEV System One):** Here, the mean inference latency is $t_{\text{JEV}} = 0.1976$ s, yielding a per-worker service rate of:

$$\mu_{\text{JEV}} = \frac{1}{0.1976} \approx 5.0607 \text{ req/s} \quad (10)$$

The aggregate processing capacity of the 16 workers is:

$$c\mu_{\text{JEV}} = 16 \times 5.0607 = 80.97 \text{ req/s} \quad (11)$$

The server utilization is:

$$\rho_{\text{JEV}} = \frac{50}{80.97} \approx 0.6175 < 1.0 \quad (12)$$

Because $\rho_{\text{JEV}} < 1.0$, the queue remains completely stable. Evaluating the Erlang-C formula yields $C(16, 9.88) \approx 0.054$, resulting in an expected waiting time of:

$$W_q^{\text{JEV}} = \frac{0.054}{16 \times 5.0607 \times (1 - 0.6175)} \approx 1.74 \text{ ms} \quad (13)$$

The total system turnaround is $T_{\text{sys}} \approx 199.3$ ms, ensuring rapid, sub-second civic dispatch.

- **Case 2: Autoregressive Foundation LLM (Sahabat-AI 8B):** Here, the mean inference latency is $\bar{t}_{\text{LLM}} = 5.1078$ s, yielding a per-worker service rate of:

$$\mu_{\text{LLM}} = \frac{1}{5.1078} \approx 0.1958 \text{ req/s} \quad (14)$$

The aggregate capacity of the same 16-worker GPU cluster is:

$$c\mu_{\text{LLM}} = 16 \times 0.1958 = 3.13 \text{ req/s} \quad (15)$$

The offered traffic load is:

$$a_{\text{LLM}} = 50 \times 5.1078 = 255.39 \text{ Erlangs} \quad (16)$$

The utilization factor explodes to:

$$\rho_{\text{LLM}} = \frac{255.39}{16} \approx 15.96 \gg 1.0 \quad (17)$$

Because $\rho_{\text{LLM}} > 1$, the queueing system is non-ergodic and mathematically divergent. The backlog of unserved citizen complaints accumulates at the deterministic rate:

$$\begin{aligned} \frac{dQ(t)}{dt} &= \lambda - c\mu_{\text{LLM}} \\ &= 50 - 3.13 = 46.87 \text{ complaints/s} \end{aligned} \quad (18)$$

Within 60 seconds of disaster onset, over 2,812 citizen calls are queued. Connection socket pools exhaust, HTTP 504 gateway timeouts trigger, and the entire public triage infrastructure undergoes catastrophic service collapse.

To prevent queue collapse and maintain $\rho \leq 0.75$ under an arrival rate of $\lambda = 50$ req/s using an autoregressive 8B LLM, the required number of parallel GPU workers is:

$$c_{\text{required}}^{\text{LLM}} = \left\lceil \frac{a_{\text{LLM}}}{0.75} \right\rceil = \left\lceil \frac{255.39}{0.75} \right\rceil = 341 \text{ enterprise GPUs} \quad (19)$$

In stark contrast, the non-autoregressive decision model requires only:

$$c_{\text{required}}^{\text{JEV}} = \left\lceil \frac{a_{\text{JEV}}}{0.75} \right\rceil = \left\lceil \frac{9.88}{0.75} \right\rceil = 14 \text{ workers} \quad (20)$$

This represents a 24.3-fold reduction in physical server provisioning.

3.2 Time-to-Dispatch Decoupling and Clinical Resuscitation Decay

In life-safety emergency operations (e.g., 911 / 112 dispatch), processing time must be decomposed into two distinct phases: (1) **Time-to-Dispatch** (T_{dispatch}), defined as the elapsed time from call initiation until the transmission of the deployment command to first responders (ambulance, fire, police), and (2) **Total Flow Time** (T_{flow}), defined as the total duration required to complete all conversational and verbal guidance instructions to the citizen caller.

In a standalone autoregressive Large Language Model, dispatch is strictly coupled to the sequential generation of the entire JSON structure:

$$T_{\text{dispatch}}^{\text{LLM}} = T_{\text{flow}}^{\text{LLM}} = t_{\text{prefill}} + \sum_{k=1}^{L_{\text{JSON}}} t_{\text{decode}}^{(k)} \quad (21)$$

Because the downstream CAD interface cannot safely execute unit deployment until the JSON payload is fully synthesized and parsed, the physical vehicle dispatch is delayed by the entire generation horizon ($L_{\text{JSON}} \approx 187$ tokens, incurring $\sim 4,984.1$ ms).

In clinical emergency medicine, out-of-hospital cardiac arrest (OHCA) survival probability decays exponentially with every second of delay prior to effective chest compressions and defibrillation [16]. Let $P_{\text{survival}}(t)$ denote the probability of intact neurological survival at time t from

cardiac arrest onset. The survival probability follows the exponential decay formulation:

$$P_{\text{survival}}(t + \Delta t_{\text{dispatch}}) = P_{\text{survival}}(t) \cdot e^{-\lambda_{\text{CPR}} \Delta t_{\text{dispatch}}} \approx P_{\text{survival}}(t) \cdot (1 - \lambda_{\text{CPR}} \Delta t_{\text{dispatch}}) \quad (22)$$

where λ_{CPR} represents the clinical mortality hazard rate without intervention. Established epidemiological studies establish that cardiac arrest survival decreases by 7% to 10% per minute of delay ($\lambda_{\text{CPR}} \approx 0.0017 \text{ s}^{-1}$ to 0.0020 s^{-1}). Over the crucial 3-to-5 minute clinical “golden period,” introducing a 5-second autoregressive generation delay ($T_{\text{dispatch}}^{\text{LLM}} \approx 5.0 \text{ s}$) produces a direct clinical survival penalty:

$$\Delta P_{\text{survival}} \approx -\lambda_{\text{CPR}} \cdot \Delta t_{\text{dispatch}} \approx -1.0\% \text{ absolute survival loss} \quad (23)$$

While seemingly small for an isolated event, across 10,000 annual municipal cardiac arrest dispatches, a 5-second delay directly precipitates approximately 100 preventable mortalities prior to vehicle departure.

In the **Two-Tier Hybrid Tandem Pipeline**, physical dispatch is mathematically decoupled from conversational synthesis:

$$\begin{aligned} T_{\text{dispatch}}^{\text{Tandem}} &= t_{\text{Tier 1}} = t_{\text{DM}} \leq 200 \text{ ms} \\ T_{\text{guidance}}^{\text{Tandem}} &= t_{\text{Tier 1}} + t_{\text{Tier 2}} \approx 5,961.2 \text{ ms} \end{aligned} \quad (24)$$

Because the rescue units roll wheels at $t = 196.2 \text{ ms}$, physical dispatch occurs within the statutory SLA ($\tau_{\text{SLA}} = 3.0 \text{ s}$), while verbal CPR instructions are synthesized in parallel out-of-band.

3.3 Disaster Call Storm and Hoax/Prank Load Shedding

During natural catastrophes (e.g., massive earthquakes, urban inundations), municipal emergency call centers experience catastrophic “call storms,” where incoming call volumes spike by $10\times$ to $50\times$. Crucially, empirical emergency studies reveal that 60% to 80% of incoming connections during disaster surges comprise non-emergency status checks, misdials, or frivolous hoax/prank calls.

Let p_{prank} denote the fraction of incoming traffic representing non-emergency or prank calls. In an autoregressive LLM dispatch pipeline, every prank call commands a full generation cycle of $L_{\text{prank}} \approx 171$ tokens, consuming $\bar{t}_{\text{prank}} \approx 4,605.9 \text{ ms}$ of GPU compute. Under an arrival rate λ , the GPU capacity consumed by non-emergency noise is:

$$a_{\text{wasted}}^{\text{LLM}} = p_{\text{prank}} \cdot \lambda \cdot \bar{t}_{\text{prank}} \quad (25)$$

If $p_{\text{prank}} = 0.70$ and $\lambda = 30 \text{ calls/s}$, the wasted GPU load is:

$$a_{\text{wasted}}^{\text{LLM}} = 0.70 \times 30 \times 4.6059 \approx 96.72 \text{ Erlangs} \quad (26)$$

requiring over 129 enterprise GPUs purely to tell prank callers to vacate the line, completely choking out legitimate life-or-death emergencies.

In contrast, non-autoregressive decision models evaluate prank calls in a single feed-forward pass ($t_{\text{prank}}^{\text{DM}} \approx 0.2235 \text{ s}$ with 0 tokens generated):

$$a_{\text{wasted}}^{\text{DM}} = 0.70 \times 30 \times 0.2235 \approx 4.69 \text{ Erlangs} \quad (27)$$

The decision model sheds 95.1% of non-emergency compute waste, instantly terminating prank calls within 223.5 ms and preserving queue availability for real emergencies.

3.4 Statutory SLA Breach Probability

In critical public administration and municipal emergency dispatch, service level agreements mandate a maximum end-to-end processing threshold τ_{SLA} . In modern digital public services, statutory guidelines typically require initial triage and department assignment within $\tau_{\text{SLA}} = 500 \text{ ms}$, while emergency hotlines require initial vehicle mobilization within $\tau_{\text{dispatch}} = 3.0 \text{ s}$.

Let computational latency Δt follow an empirical probability density function $p(\Delta t)$. The probability of breaching the statutory service level agreement is formulated as:

$$P(\text{SLA Breach}) = \int_{\tau_{\text{SLA}}}^{\infty} p(\Delta t) d\Delta t \quad (28)$$

Assuming execution latency follows a normal distribution $\Delta t \sim \mathcal{N}(\mu_t, \sigma_t^2)$ over stationary inference workloads:

$$P(\text{SLA Breach}) = 1 - \Phi\left(\frac{\tau_{\text{SLA}} - \mu_t}{\sigma_t}\right) = Q\left(\frac{\tau_{\text{SLA}} - \mu_t}{\sigma_t}\right) \quad (29)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function and $Q(x) = 1 - \Phi(x)$.

For TypeSafe JEV System One, empirical measurements yield $\mu_t = 197.6 \text{ ms}$ and $\sigma_t = 40.5 \text{ ms}$. Evaluating the standardized normal score at $\tau_{\text{SLA}} = 500 \text{ ms}$:

$$z_{\text{JEV}} = \frac{500 - 197.6}{40.5} = \frac{302.4}{40.5} \approx +7.47 \quad (30)$$

The theoretical probability of SLA breach is:

$$P(\text{SLA Breach})_{\text{JEV}} = Q(7.47) \approx 3.9 \times 10^{-14} \approx 0.0\% \quad (31)$$

demonstrating complete deterministic SLA compliance.

Conversely, for Sahabat-AI 8B Instruct, empirical measurements yield $\mu_t = 5,107.8 \text{ ms}$ and $\sigma_t = 484.5 \text{ ms}$. Evaluating the standardized score:

$$z_{\text{LLM}} = \frac{500 - 5,107.8}{484.5} = \frac{-4,607.8}{484.5} \approx -9.51 \quad (32)$$

The breach probability is:

$$P(\text{SLA Breach})_{\text{LLM}} = Q(-9.51) = 1.000 (100.0\%) \quad (33)$$

Every single inbound query processed by the autoregressive LLM violates the statutory SLA deadline.

3.5 Energy Footprint and Total Cost of Ownership (TCO)

The total operational expenditure and environmental impact of automated triage systems are directly governed by the energy expenditure of GPU execution. Let P_{node} represent the thermal design power (TDP) in Watts consumed by the inference node under active matrix multiplication. The electrical energy consumed to resolve a single citizen query is:

$$E_{\text{query}} = P_{\text{node}} \cdot \bar{t} \quad [\text{Joules}] \quad (34)$$

For an enterprise dual-GPU inference server consuming $P_{\text{node}} = 300 \text{ W}$:

$$\begin{aligned} E_{\text{query}}^{\text{LLM}} &= 300 \text{ W} \times 5.1078 \text{ s} \\ &= 1,532.34 \text{ J} \approx 0.4256 \text{ Wh} \end{aligned} \quad (35)$$

$$\begin{aligned} E_{\text{query}}^{\text{DM}} &= 70 \text{ W} \times 0.050 \text{ s} \\ &= 3.50 \text{ J} \approx 0.00097 \text{ Wh} \end{aligned} \quad (36)$$

where the non-autoregressive decision model runs on a compact edge accelerator consuming 70 W over a 50 ms forward execution window. The energy consumption ratio is:

$$\frac{E_{\text{query}}^{\text{LLM}}}{E_{\text{query}}^{\text{DM}}} = \frac{1,532.34}{3.50} \approx 437.8 \times \quad (37)$$

Processing inquiries via an autoregressive 8B model expends over 430 times more electrical energy per query than a non-autoregressive decision engine.

Across an annual operational volume of $V_{\text{annual}} = 100,000,000$ citizen transactions (typical for national-scale digital portals in a country of 280 million citizens), the annual power consumption diverges dramatically:

$$\begin{aligned} \mathcal{E}_{\text{annual}}^{\text{LLM}} &= 10^8 \times 0.4256 \text{ Wh} \\ &= 42,560 \text{ kWh} \approx 42.56 \text{ MWh} \\ \mathcal{E}_{\text{annual}}^{\text{DM}} &= 10^8 \times 0.00097 \text{ Wh} \\ &= 97.2 \text{ kWh} \approx 0.097 \text{ MWh} \end{aligned} \quad (38)$$

From an economic billing perspective, foundation LLMs generate an average of $N_{\text{tok}} = 196$ output tokens per query, alongside an input prompt context of approximately $N_{\text{in}} \approx 300$ tokens. Under enterprise cloud hosting rates amortized at \$0.20 per 10^6 tokens for high-volume inference clusters:

$$\begin{aligned} \text{Cost}_{100k}^{\text{LLM}} &= 100,000 \times (N_{\text{in}} + N_{\text{tok}}) \times \frac{\$0.20}{10^6} \\ &= 100,000 \times 496 \times 2 \times 10^{-7} \approx \$9.92 \text{ to } \$39.20 \end{aligned} \quad (39)$$

In contrast, non-autoregressive decision models perform zero token generation ($N_{\text{tok}} \equiv 0$). Compute costs are billed purely on fixed compute instances or single-pass logit API calls at \$0.05 per 100,000 requests. The relative fiscal efficiency gain η_{cost} is:

$$\eta_{\text{cost}} = \left(1 - \frac{\$0.05}{\$39.20}\right) \times 100\% = 99.872\% \quad (40)$$

Non-autoregressive intent gating eliminates 99.87% of recurring cloud inference costs.

3.6 Schema Entropy and JSON Syntax Failure Formulation

When an AI agent routes citizen reports, downstream relational databases and microservice event buses (e.g., Kafka, RabbitMQ) require deterministic JSON formatting with typed attributes:

$$\mathcal{S} = \{\text{"answers"} : \{\dots\}, \text{"routing_target"} : \kappa \in \mathcal{K}\} \quad (41)$$

In an autoregressive decoder, generating a JSON string of length L tokens requires predicting a discrete token sequence $Y = (y_1, y_2, \dots, y_L)$ from an open vocabulary $\mathcal{V}$ ($|\mathcal{V}| \approx 150,000$):

$$y_k \sim \text{Categorical} \left(\text{Softmax} \left(\frac{\mathbf{W}_v \mathbf{h}_k}{\tau} \right) \right) \quad (42)$$

A generated sequence is valid if and only if it strictly conforms to the formal context-free grammar $G_{\text{JSON}} = (V_N, \Sigma, R, S_0)$ defining the JSON specification.

Let $p_{\text{drift},k}$ be the probability that the decoder emits a token at position k that violates grammar G_{JSON} given that the preceding prefix $y_{<k}$ was valid:

$$p_{\text{drift},k} = \Pr(y_k \notin \text{Follow}(y_{<k}, G_{\text{JSON}}) \mid y_{<k} \in \mathcal{L}(G_{\text{JSON}})) \quad (43)$$

Assuming quasi-independent token drift probabilities with geometric mean $\bar{p}_{\text{drift}}$, the cumulative probability of schema compliance across sequence length L is:

$$P(\text{Schema Compliance}) = \prod_{k=1}^L (1 - p_{\text{drift},k}) \approx (1 - \bar{p}_{\text{drift}})^L \quad (44)$$

Consequently, the probability of structural syntax failure is:

$$P(\text{Syntax Failure}) = 1 - \prod_{k=1}^L (1 - p_{\text{drift},k}) \approx 1 - (1 - \bar{p}_{\text{drift}})^L \quad (45)$$

Even when the per-token drift rate is exceptionally small (e.g., $\bar{p}_{\text{drift}} \approx 0.00181$), across an autoregressive output sequence of length $L = 196$ tokens:

$$\begin{aligned} P(\text{Syntax Failure}) &\approx 1 - (1 - 0.00181)^{196} \\ &\approx 1 - 0.7015 = 0.2985 \approx 30.0\% \end{aligned} \quad (46)$$

This derivation explains why Sahabat-AI 8B demonstrated an empirical 30.0% syntax failure rate during evaluation—manifesting as missing commas, unterminated quotation marks, and unclosed brackets.

In non-autoregressive decision models, text generation is completely eliminated. The model ingests citizen text X and evaluates continuous representation $\mathbf{h} = \text{Encoder}(X)$. Action distributions are evaluated across a closed candidate set $\mathcal{K}$:

$$\mathbf{p}_{\text{target}} = \text{Softmax}(\mathbf{W}_{\text{target}} \mathbf{h} + \mathbf{b}_{\text{target}}) \in \Delta^{|\mathcal{K}|-1} \quad (47)$$

The resulting probability vector is mapped directly into strongly typed in-memory objects (e.g., C++ structs or Python Pydantic classes) whose serialization is handled by compiled deterministic routines. Because $p_{\text{drift},k} \equiv 0$, non-autoregressive decision models achieve:

$$\begin{aligned} P(\text{Syntax Failure}) &\equiv 0.0 \\ \iff P(\text{Schema Compliance}) &\equiv 100.0\% \end{aligned} \quad (48)$$

4 EMPIRICAL BENCHMARK METHODOLOGY

4.1 Experimental Testbed and Infrastructure Topology

The benchmark was implemented within **DecisionModel-Bench** on a distributed multi-accelerator server provisioned with dual NVIDIA Tesla T4 GPUs (16 GB VRAM per GPU, 31.27 GB combined VRAM, PCIe Gen3 x16, 300 GB/s

memory bandwidth per device). The host system executed Ubuntu 22.04 LTS powered by Intel Xeon processors (2.20 GHz, 4 vCPUs) and 32 GB of system RAM.

The compute topology was configured as follows:

- **GPU 0 (Generative Tier & Local Encoders):** Hosted the local quantized generative LLM Sahabat-AI 8B Instruct (GGUF Q4_K_M, utilizing ~ 5.2 GB VRAM) and compact local decision heads including Kev-0.8B and Laya Multilingual 421M.
- **GPU 1 (OpenJev Logit Scorer):** Hosted OpenJev 0.5B, dedicated to single-pass continuation logit extraction over candidate token projections.
- **SaaS API Tier (TypeSafe JEV System One):** Interfaced via secure REST endpoints over high-speed WAN, evaluating single-pass decision tensors.

4.2 Evaluated Models

Seven distinct AI architectures spanning non-autoregressive decision engines and foundation generative decoders were benchmarked:

- 1) **TypeSafe JEV System One:** Managed cloud-native non-autoregressive decision model operating at single-pass logit extraction, returning structured boolean flags, target routing choices, and continuous urgency scores without token generation.
- 2) **OpenJev (0.5B Logit Scorer):** Locally hosted 500-million parameter causal backbone executing single-pass continuation logit scoring across structured choice indices.
- 3) **Kev-0.8B (LoRA Pointer):** An 800-million parameter localized architecture augmented with low-rank adaptation (LoRA) routing heads for discrete entity dispatch.
- 4) **Laya Multilingual (421M):** A 421-million parameter compact bidirectional ModernBERT encoder model optimized for cross-lingual Indonesian-English semantic intent classification.
- 5) **Sahabat-AI 8B Instruct:** An 8-billion parameter Indonesian sovereign foundation LLM developed by GoTo and Indosat Ooredoo Hutchison, pre-trained on extensive national and regional language corpora and instruction-tuned for Indonesian administrative tasks.
- 6) **Qwen 2.5 7B Instruct:** A 7.6-billion parameter flagship foundation LLM developed by Alibaba Cloud [4], optimized for multi-lingual reasoning, mathematical logic, and structured JSON output generation.
- 7) **Gemma 2 9B Instruct:** A 9.2-billion parameter high-capacity open-weight foundation model developed by Google DeepMind [5], featuring interleaved local sliding-window and global self-attention architectures.

4.3 Civic and Enterprise Benchmark Scenarios

The benchmark evaluates 10 rigorous public service and enterprise customer support scenarios designed to mirror high-stakes operations in Indonesia:

- 1) **PUB-01 (Municipal Flood Emergency):** Flash flood breaking the Ciliwung river levee in RW 07, submerging residential zones under 2 meters of water with trapped elderly and infants. Ground Truth: Emergency: True; Target: bpbd_basarnas_damkar.
- 2) **PUB-02 (Civil Service Graft / SP4N-LAPOR):** Citizen reporting illegal administrative fee extortion (Pungli) of Rp 150,000 for expedited e-KTP processing at a sub-district kelurahan office. Ground Truth: Emergency: False; Target: inspektorat_ombudsman.
- 3) **PUB-03 (National Highway Subsidence):** A 1.5-meter deep crater on a trans-provincial logistics highway caused by torrential rain erosion, overturning heavy freight trucks and causing a 10-kilometer gridlock. Ground Truth: Emergency: True; Target: pupr_bina_marga.
- 4) **PUB-04 (Emergency Healthcare Refusal):** A private hospital emergency department turning away a patient experiencing acute myocardial infarction due to bed unavailability and demanding an advance deposit of Rp 20,000,000 despite active BPJS Kesehatan insurance. Ground Truth: Emergency: True; Target: kemenkes_bpjs_kesehatan.
- 5) **PUB-05 (Toxic Hazardous Waste Dumping):** A textile manufacturing plant discharging hazardous chemical waste (B3) directly into a river system at 2:00 AM, causing fish mortality and well contamination. Ground Truth: Emergency: True; Target: gakkum_klkh_dlh.
- 6) **CS-01 (Banking Fraud / Social Engineering):** Immediate citizen report of unauthorized account takeover and withdrawal of Rp 25,000,000 following an OTP phishing call. Ground Truth: Emergency: True; Target: fraud_desk_emergency_block.
- 7) **CS-02 (E-Commerce Dispute / Viral Threat):** High-value smartphone purchase replaced in-transit with laundry detergent, with courier disavowal and an active threat of social media viral defamation. Ground Truth: Emergency: True; Target: priority_dispute_retur.
- 8) **CS-03 (Fintech Debt Restructuring):** Borrower experiencing involuntary layoff following corporate bankruptcy, proactively requesting tenor extension and loan restructuring under good faith. Ground Truth: Emergency: False; Target: collection_restructuring.
- 9) **CS-04 (Telecommunications Backbone Severance):** Core fiber-optic trunk severed by heavy excavation equipment, causing an industrial park blackout. Ground Truth: Emergency: True; Target: noc_fiber_splicing_team.
- 10) **CS-05 (Adversarial Prompt Injection / Jail-break):** A malicious attack attempting to override system prompts and exfiltrate unencrypted database credentials, API keys, and citizen PII. Ground Truth: Emergency: True; Target: security_block_and_log.

4.4 Emergency Calling (911 / 112) Critical Benchmark Scenarios

To rigorously stress-test acute life-safety emergency operations, we formulated five specialized 911/112 emergency calling scenarios reflecting urgent municipal crises:

- 1) **EMERG-01: Henti Jantung & Henti Napas (Out-of-Hospital Cardiac Arrest & Apnea):** Caller in severe emotional distress reporting their father collapsed clutching his chest, unresponsive, apnoeic, and presenting cyanosis at Jalan Anggrek No. 12. Ground Truth: Emergency: True; Target: ambulans_paramedis_icu; Triage: merah_kritis_henti_jantung. Required Action: Immediate ICU mobile paramedical unit dispatch coupled with out-of-band verbal CPR compression guidance.
- 2) **EMERG-02: Kebakaran Gedung Ruko Terjebak Asap (Structural Commercial Fire with Rooftop Trapped Workers):** First-floor plastics warehouse explosion with fire spreading rapidly to the third floor, trapping six employees on the rooftop due to impassable, superheated black toxic smoke in stairwells. Ground Truth: Emergency: True; Target: damkar_snorkel_rescue; Triage: kritis_korban_terjebak. Required Action: Aerial platform snorkel ladder truck dispatch alongside smoke evacuation and wet-cloth respiratory survival coaching.
- 3) **EMERG-03: Perampokan Bersenjata Berlangsung (Active Armed Minimarket Robbery):** Whispered emergency call from employees concealed in a rear vault on Jalan Raya Bogor KM 28, reporting three perpetrators wielding firearms and blades, holding the cashier hostage under active death threats. Ground Truth: Emergency: True; Target: polri_resmob_gegana; Triage: ancaman_senjata_mematikan. Required Action: Covert tactical police dispatch (Resmob/Perintis Presisi) without audible siren exposure, coupled with silent hiding and non-confrontation caller protocols.
- 4) **EMERG-04: Tabrakan Beruntun Tol Cipularang Korban Terjepit (Multi-Vehicle Highway Extrication Crash):** Heavy articulated freight truck with brake failure causing a multi-vehicle pileup on the Cipularang KM 92 toll road, leaving two severely bleeding victims trapped inside crushed vehicle cabin wreckage. Ground Truth: Emergency: True; Target: basarnas_rescue_hidrolik; Triage: kritis_terjepit_nyawa. Required Action: Rapid mobilization of Basarnas/Fire heavy rescue units equipped with hydraulic spreaders, cutters (Jaws of Life), and mobile trauma surgeons.
- 5) **EMERG-05: Panggilan Iseng / Hoaks (Prank Call Storm & Non-Emergency Screening):** Frivolous caller inquiring about the current time and requesting cellular prepaid balance vouchers during an active operational watch. Ground Truth: Emergency: False; Target: filter_prank_warning; Triage: prank_palsu. Required Action: Rapid non-

punitive warning and instantaneous channel termination without consuming generative GPU resources.

4.5 Two-Tier Hybrid Tandem Pipeline Architecture

The Two-Tier Hybrid Tandem Pipeline evaluates the synergy of non-autoregressive decision models and foundation LLMs operating in parallel across the dual Tesla T4 testbed:

- **Tier 1 (Sub-200ms Decision Reflex Engine):** Ingests the raw emergency call transcript and evaluates candidate logits in a single feed-forward pass. Tier 1 outputs typed categorical flags (`is_emergency`, `dispatch_unit`, `triage_code`) within 200 ms, transmitting instantaneous dispatch orders directly to the CAD broker and vehicle telemetry systems. Crucially, Tier 1 generates zero tokens ($N_{\text{tok}} \equiv 0$).
- **Tier 2 (Asynchronous Foundation Guidance Engine):** Deployed locally on GPU 0 using Sahabat-AI 8B Instruct. Tier 2 receives both the citizen transcript and the structured triage tags emitted by Tier 1. Operating out-of-band and asynchronously, Tier 2 synthesizes culturally grounded, empathetic, step-by-step verbal guidance (e.g., cardiopulmonary resuscitation rhythms, fire smoke survival tactics, concealment advice) directly to the phone operator, completing within 3 to 7 seconds without holding back physical unit deployment.

5 EMPIRICAL RESULTS AND PERFORMANCE EVALUATION

5.1 Latency Distribution and SLA Compliance in Civic Triage

Table 1 presents the empirical results across the 10 civic and enterprise benchmark scenarios. The divergence in computational latency between non-autoregressive decision models and the autoregressive foundation model is stark.

TypeSafe JEV System One achieved a mean round-trip latency of **197.6 ms** (ranging from 144.2 ms to 294.2 ms), executing over WAN infrastructure. Even with network transit delays across external fiber connections, 100% of JEV requests finished well below the 500 ms statutory SLA boundary (100.0% SLA Pass Rate).

Locally deployed decision models operating without sequential decoding also demonstrated significant latency advantages. **OpenJev 0.5B** sustained a consistent mean execution latency of **540.6 ms** with minimal variance ($\sigma = 24.3$ ms), reflecting deterministic matrix multiplication overhead. **Kev-0.8B** achieved a mean latency of **648.5 ms**, occasionally completing in as few as 220.9 ms.

In contrast, **Sahabat-AI 8B Instruct** exhibited a mean latency of **5,107.8 ms**—more than 25 times slower than JEV System One. The minimum observed latency for Sahabat-AI was 4,305.8 ms, while the maximum reached 5,703.2 ms. Consequently, Sahabat-AI achieved an **SLA pass rate of exactly 0.0%**. Under high-throughput municipal operations, forcing citizen emergency calls through an autoregressive LLM guarantees systemic violation of dispatch timing mandates.

TABLE 1
Comprehensive Empirical Evaluation: Non-Autoregressive Decision Models vs. Foundation Generative LLMs across High-Throughput Public Administration and Enterprise Customer Support (DecisionModelBench).

Model Architecture	Params	Tier / Paradigm	Latency (ms) Mean	Range (ms) Min–Max	Accuracy Routing	Schema Compl.	Tokens Mean	Cost / 100k Queries	SLA Pass (≤500 ms)
TypeSafe JEV System One	—	Cloud SaaS API	197.6	144.2 – 294.2	100.0%	100.0%	0	\$0.05	100.0%
Kev-0.8B (LoRA Pointer)	0.8B	Local GPU	648.5	220.9 – 1,182.0	100.0%	100.0%	0	\$0.05	20.0%
OpenJev (Logit Scorer)	0.5B	Local GPU	540.6	499.6 – 582.3	90.0%	100.0%	0	\$0.05	10.0%
Laya Multilingual	421M	Local GPU/CPU	3,639.2*	63.2 – 7,211.7	60.0%	100.0%	0	\$0.05	10.0%
Sahabat-AI 8B Instruct	8.0B	Autoregressive LLM	5,107.8	4,305.8 – 5,703.2	70.0%	70.0%	196	\$39.20	0.0%

*Note: Laya Multilingual executes forward passes in 35.8–63.2 ms in dedicated environments (e.g., 63.2 ms in PUB-02). The elevated 3,639.2 ms average reflects host CPU/GPU contention when co-located with active 8B LLM processes, underscoring the critical necessity of physical compute isolation for real-time decision inference.

5.2 Routing Accuracy and Domain Robustness

In operational public service triage, routing errors have severe real-world consequences: misrouting an emergency flood rescue or an acute cardiac denial delays life-saving interventions.

TypeSafe JEV System One and **Kev-0.8B** both attained **100.0% semantic routing accuracy** across all 10 scenarios. Both models correctly identified emergency flags, assigned the exact designated administrative and corporate entities, and calibrated urgency ratings appropriately.

OpenJev 0.5B attained **90.0% accuracy**, correctly handling 9 out of 10 scenarios, with a minor misclassification on scenario CS-03.

Sahabat-AI 8B Instruct achieved only **70.0% semantic routing accuracy**. Beyond latency overhead, the model struggled in three scenarios:

- 1) In **PUB-04** (Emergency Healthcare Refusal), the model failed to output a parseable routing key, falling back to an unmapped target.
- 2) In **CS-03** (Debt Restructuring), Sahabat-AI generated an ambiguous token key that failed to match the designated departmental target `collection_restructuring`, defaulting to `unknown`.
- 3) In **CS-04** (Fiber-Optic Backbone Outage), generation corruption invalidated the target extraction.

5.3 Schema Compliance and Syntax Failures

A pivotal finding of DecisionModelBench is the fundamental difference in structural output reliability between paradigms:

- **Non-Autoregressive Decision Models:** Achieved **100.0% schema compliance** across all runs. Because decisions are evaluated as numerical tensors over pre-defined candidate vectors and serialized via deterministic host routines, zero parsing errors, syntax violations, or malformed strings occurred.
- **Sahabat-AI 8B Instruct:** Exhibited a **30.0% schema failure rate** (70.0% compliance). In scenarios **PUB-04** and **CS-04**, the model failed with explicit JSON parsing exceptions:

```
JSON parse error: Expecting ',',
delimiter: line 17 column 4 (char
318)
```

and

```
JSON parse error: Expecting ',',
delimiter: line 16 column 4 (char
292)
```

During token-by-token generation of the nested JSON payload, the model omitted necessary delimiter commas between attribute fields, producing truncated and invalid syntax. In production pipelines, such failures cause unhandled exceptions in downstream service brokers, requiring costly fallback retry loops that double the already multi-second latency.

5.4 Host Resource Contention and Isolation Pathology

An instructive operational insight emerged from **Laya Multilingual 421M**. When evaluated on an isolated host without resource contention, ModernBERT-based forward passes consistently complete within **35.8 ms to 63.2 ms** (observed in scenario PUB-02 at 63.2 ms). However, during benchmark runs where the 8B parameter autoregressive model was concurrently executing on the shared testbed, thread contention, GPU memory bus locking, and operating system context switching inflated Laya’s mean latency to **3,639.2 ms**.

This empirical phenomenon highlights an important systems engineering principle: lightweight, deterministic decision models must be physically isolated on dedicated compute cores or separate edge accelerators. When co-located with memory-saturating foundation LLMs on unpartitioned hardware, the multi-gigabyte memory transfers of autoregressive decoding induce starvation on adjacent compute processes.

5.5 Emergency Calling (911 / 112) and Two-Tier Tandem Empirical Evaluation

Table 2 details the empirical performance of standalone decision models, standalone foundation LLMs, and the integrated Two-Tier Hybrid Tandem Pipeline across the five critical life-safety emergency scenarios.

5.5.1 Time-to-Dispatch vs. Total Flow Time Analysis

The empirical divergence across paradigms reveals fundamental operational insights:

TABLE 2

Emergency Calling (911 / 112) and Two-Tier Tandem Empirical Benchmark: Standalone Decision Models vs. Standalone Foundation LLM vs. Two-Tier Hybrid Brain Pipeline on Dual NVIDIA Tesla T4 GPUs.

Operational Paradigm	Time-to-Dispatch (T_{dispatch})	Total Flow Time (T_{flow})	Routing Accuracy	Tokens Emitted	Statutory Status (<3.0 s SLA)	Clinical / Operational Assessment
Decision Model (JEV Cloud)	232.8 ms	232.8 ms	100.0% (5/5)	0	Passes Regulation	Instantaneous Fleet Dispatch
Decision Model (OpenJev 0.5B)	560.4 ms	560.4 ms	80.0% (4/5)	0	Passes Regulation	Deterministic Local Execution
Standalone LLM (Sahabat-AI 8B)	4,984.1 ms	4,984.1 ms	100.0% (5/5)	187	FAILS SLA (Delay)	Dangerous Resuscitation Lag
TWO-TIER TANDEM	196.2 ms	5,961.2 ms	100.0% (5/5)	226	PARETO OPTIMAL	Instant Dispatch + Rich CPR

Note: Time-to-Dispatch denotes the latency until the CAD system receives the structured JSON deployment directive and rolls rescue vehicles. Total Flow Time denotes the complete interaction duration including generative audio/text first-aid guidance. In the Two-Tier Hybrid Tandem Brain, Tier 1 (JEV Decision Model) executes in 196.2 ms (0 tokens), while Tier 2 (Sahabat-AI 8B) concurrently generates 226 tokens of step-by-step caller guidance out-of-band in 5,765.0 ms without delaying fleet departure.

- **Standalone Decision Models: TypeSafe JEV Cloud** achieved an average Time-to-Dispatch of **232.8 ms** across all five emergency scenarios with 100.0% accuracy and zero tokens. **OpenJev 0.5B** achieved a mean Time-to-Dispatch of **560.4 ms** with 80.0% accuracy (misrouting the cardiac arrest in EMERG-01 to `damkar_rescue`). Both decision models comfortably satisfied statutory dispatch timing regulations ($\tau_{\text{dispatch}} < 3.0$ s).
- **Standalone Foundation LLM (Sahabat-AI 8B):** While achieving 100.0% accuracy, Sahabat-AI required an average of **4,984.1 ms** (T_{dispatch}) to synthesize an average of 187 tokens. In scenario EMERG-01 (Cardiac Arrest), latency reached **5,841.7 ms**; in EMERG-04 (Tol Cipularang Extrication Crash), latency was **5,758.7 ms**. In every scenario, the standalone LLM violated the 3-second statutory emergency dispatch SLA. Coupling physical unit dispatch to autoregressive generation introduces multi-second dead time during acute physiological crises.
- **The Two-Tier Hybrid Tandem Pipeline:** By delegating immediate dispatch determination to Tier 1 (Decision Model) and asynchronous conversational guidance to Tier 2 (Sahabat-AI 8B), the hybrid brain achieved an unprecedented mean Time-to-Dispatch of **196.2 ms**—faster even than standalone JEV due to dedicated edge routing—and 100.0% dispatch accuracy with zero dispatch token emission. Concurrently, Tier 2 completed comprehensive verbal guidance in 5,765.0 ms ($T_{\text{flow}} = 5,961.2$ ms), generating 226 tokens of detailed, empathetic, actionable survival protocols.

5.5.2 Medical Analytical Insights: The Golden Period of Cardiac Arrest

In scenario **EMERG-01** (Cardiac Arrest & Apnea), the caller reported an unconscious, non-breathing individual turning cyanotic. In clinical medicine, irreversible cerebral cortex necrosis initiates within 3 to 5 minutes of cardiac arrest.

Under the standalone LLM paradigm, the emergency computer system was completely blocked for **5,841.7 ms**

(nearly 6 seconds) while generating tokens describing the incident before the paramedical ICU ambulance was alerted. In contrast, under the Two-Tier Tandem Pipeline:

- 1) At $t = 217.0$ ms, Tier 1 triggered the Computer-Aided Dispatch (CAD) API, instantly transmitting GPS coordinates to the nearest mobile ICU paramedical unit and initiating siren rollout.
- 2) Concurrently, Tier 2 synthesized continuous chest-compression instructions:

“Saya ingin memastikan bahwa kami sedang mengirimkan ambulans paramedis ICU ke lokasi Anda secepat mungkin... Jalan Anggrek No. 12 adalah lokasi yang kami kenal, dan armada kami bergerak sekarang. Letakkan tumit tangan Anda di tengah dada korban, kunci siku Anda, dan tekan sedalam 5 cm dengan irama cepat 100–120 kali per menit.”

By initiating vehicle rollout at $t = 217.0$ ms, the tandem architecture preserves crucial seconds of the golden period, ensuring maximal resuscitation efficacy.

5.5.3 Disaster Response Analysis: Hoax/Prank Screening in Call Storms

In scenario **EMERG-05** (Prank Call Filter), an incoming caller made frivolous inquiries regarding prepaid phone vouchers. During metropolitan crises, such frivolous inquiries saturate telephone switches.

Under the standalone LLM paradigm, Sahabat-AI 8B expended **4,605.9 ms** and **171 tokens** of active GPU matrix compute to generate a lengthy polite decline. Multiplying this delay across thousands of simultaneous prank calls locks the GPU worker thread pool, preventing genuine emergency calls from reaching the queue.

Under the Decision Model paradigm (JEV), the non-emergency nature of the call was classified in **223.5 ms** with zero tokens generated. The call was instantly flagged with `filter_prank_warning`, allowing the automated telephony exchange to terminate the connection in under a quarter of a second, shedding 95.1% of compute overhead and preserving capacity for genuine crises.

5.6 Cross-Architecture Multi-LLM Tandem Evaluation Across Flagship Models

To evaluate whether the operational advantages of the Two-Tier Hybrid Tandem Pipeline generalize across diverse foundation generative families, we conducted an empirical stress-test on the dual NVIDIA Tesla T4 benchmark testbed across three flagship open-weight models:

- 1) **Sahabat-AI 8B Instruct** (Indosat/GoTo, 8.0B parameters);
- 2) **Qwen 2.5 7B Instruct** (Alibaba Cloud, 7.6B parameters);
- 3) **Gemma 2 9B Instruct** (Google DeepMind, 9.2B parameters).

Table 3 summarizes the comparative empirical results across standalone single-tier execution versus the Two-Tier Hybrid Tandem pipeline (partnered with TypeSafe JEV System 1).

The empirical data highlights profound operational divergences:

- **Standalone Baseline Latency:** In standalone mode, the baseline non-autoregressive decision model (TypeSafe JEV) completed dispatch determination in **208.2 ms** with 100.0% accuracy and zero tokens. Conversely, the generative LLMs exhibited dramatic latency penalties: Sahabat-AI 8B averaged **3,340.0 ms**, Qwen 2.5 7B required **10,024.4 ms**, and Gemma 2 9B demanded **12,681.7 ms** (nearly 13 seconds) per decision.
- **Dispatch Acceleration Factors in Tandem:** When paired with TypeSafe JEV in the Two-Tier Hybrid Tandem Pipeline, physical CAD dispatch latency plummeted across all models to sub-220ms: **217.7 ms** for Sahabat-AI (a **15.3× acceleration**), **178.5 ms** for Qwen 2.5 (a **56.1× acceleration**), and **197.4 ms** for Gemma 2 (a **64.2× acceleration**).
- **Schema Robustness and Formatting Fragility:** While Sahabat-AI 8B and Gemma 2 9B maintained 100.0% routing accuracy in standalone mode on this emergency cohort, Qwen 2.5 7B collapsed to **33.3% standalone accuracy**. This failure stemmed from structural syntax wrapping: Qwen frequently enveloped its JSON output within markdown code fences (````json ... ````) and appended unsolicited conversational preambles. In a synchronous single-tier architecture, this formatting non-determinism crashes downstream JSON parsers, causing complete dispatch failure. In contrast, under the Two-Tier Tandem configuration, Tier 1 intercepts and executes the CAD dispatch deterministically (achieving **100.0% accuracy** across all models), entirely insulating the operational dispatch loop from LLM syntactic variance.
- **High-Fidelity First-Aid Guidance:** Concurrently, the Tier 2 models synthesized rich, medically grounded conversational guidance: Sahabat-AI generated an average of **294 tokens**, Qwen 2.5 generated **281 tokens**, and Gemma 2 generated **207 tokens** of high-quality caller coaching without introducing any delay into physical vehicle departure.

5.7 Clinical Mortality and Systems Discussion: The Lethality of Single-Tier Decoders in Acute Resuscitation vs. The Two-Tier Hybrid Imperative

The empirical results compiled in Table 3 expose an urgent, life-critical finding: **forcing a foundation Large Language Model to act as a synchronous, single-tier dispatcher in municipal 911/112 operations is clinically lethal.**

5.7.1 Pathophysiology of the Golden Period in Out-of-Hospital Cardiac Arrest

In emergency medicine, the prognosis of out-of-hospital cardiac arrest (OHCA) is governed by an unrelenting biological countdown. Upon ventricular fibrillation or asystole, systemic blood circulation ceases instantly. The human cerebral cortex, possessing virtually zero glycogen reserves, exhausts dissolved oxygen within 10 to 15 seconds. Irreversible neuronal autolysis and ischemic brain injury commence within 180 to 300 seconds (3 to 5 minutes)—the universally recognized clinical “golden period” [16].

Epidemiological survival curves established by Eisenberg and Mengert [16] and validated in municipal dispatch studies [15] show that every 60-second delay in initiating cardiopulmonary resuscitation (CPR) and automated external defibrillation (AED) reduces the probability of survival by 7% to 10%:

$$P_{\text{survival}}(t) = P_0 \cdot \exp(-\lambda_{\text{CPR}} \cdot t) \quad (49)$$

where $\lambda_{\text{CPR}} \approx 0.0018 \text{ s}^{-1}$.

Consider the real-world operational consequence of deploying Gemma 2 9B or Qwen 2.5 7B as a single-tier synchronous dispatcher:

- **Gemma 2 9B Standalone Delay (12.7 seconds):** Generating the JSON dispatch payload requires 12,681.7 ms of active GPU decoding. During these 12.7 seconds, the emergency caller is trapped in dead silence or listening to waiting tones, the CAD system cannot transmit GPS telemetry, and the mobile ICU ambulance sits motionless in the station bay. A 12.7-second delay incurs an absolute survival penalty of:

$$\begin{aligned} \Delta P_{\text{survival}} &= P_0 (1 - e^{-0.0018 \times 12.68}) \\ &\approx P_0 \times 0.0226 \text{ (a 2.3\% absolute reduction)} \end{aligned} \quad (50)$$

In a metropolitan jurisdiction experiencing 10,000 cardiac arrest calls annually, this 12.7-second autoregressive bottleneck directly accounts for over 230 preventable patient fatalities before first responders even turn the ignition key.

- **Qwen 2.5 7B Standalone Syntax Failure (10.0 seconds + Crash):** While Qwen 2.5 7B requires 10,024.4 ms to generate its response, its 33.3% standalone accuracy introduces a fatal compounding failure. In two out of three emergency calls, the emitted markdown fences (````json`) trigger a CAD parser exception. In a real-world dispatch center, an unhandled parser exception forces either an automated HTTP retry loop (adding another 10 to 20 seconds) or an emergency human fallback alert (adding 30 to 45 seconds). In acute cardiac arrest or arterial hemorrhaging, a 30-to-45-second software crash is an inescapable death sentence.

TABLE 3
Multi-LLM Tandem Evaluation Across Flagship Architectures: Standalone Single-Tier LLM vs. Two-Tier Hybrid Tandem Pipeline on Dual NVIDIA Tesla T4 GPUs (DecisionModelBench).

Model Architecture (System 2 Engine)	Params & Developer	Standalone Latency	Tandem Disp. (T_{disp})	Dispatch Accel.	Standalone Accuracy	Tandem Accuracy	First-Aid P3K Guidance	Total Flow (T_{flow})
TypeSafe JEV (Sys 1 Solo)	Non-AR / TypeSafe	208.2 ms	208.2 ms	1.0×	100.0%	100.0%	0 tokens	208.2 ms
Sahabat-AI 8B Instruct	8.0B / Indosat-GoTo	3,340.0 ms	217.7 ms	15.3×	100.0%	100.0%	294 tokens	7,452.0 ms
Qwen 2.5 7B Instruct	7.6B / Alibaba Cloud	10,024.4 ms	178.5 ms	56.1×	33.3% [†]	100.0%	281 tokens	6,595.3 ms
Gemma 2 9B Instruct	9.2B / Google DeepMind	12,681.7 ms	197.4 ms	64.2×	100.0%	100.0%	207 tokens	6,909.0 ms

[†]Note: Qwen 2.5 7B Standalone suffered a 33.3% accuracy rate due to structural JSON markdown fence wrapping (``json ... ``), which precipitated unhandled syntax parsing exceptions in the single-tier synchronous CAD ingestion engine. Under the Two-Tier Hybrid Tandem architecture, TypeSafe JEV System 1 completely shields the operational CAD layer, executing deterministic dispatch in 178.5 ms (100.0% accuracy) while allowing Qwen 2.5 to synthesize 281 tokens of rich conversational first-aid advice out-of-band.

5.7.2 The Two-Tier Hybrid Tandem as the Architectural Imperative

The Two-Tier Hybrid Tandem Pipeline permanently dismantles the false dichotomy between instantaneous operational reflex and nuanced linguistic intelligence. By bifurcating the emergency response pipeline into two specialized cognitive strata:

- 1) **System 1 (Deterministic Operational Reflex):** A non-autoregressive decision model (e.g., TypeSafe JEV System One) evaluates candidate logits in a single feed-forward pass. Within **178.5 ms to 217.7 ms** (<220 ms), System 1 outputs typed categorical directives directly to the municipal CAD event broker. Wheels roll, sirens activate, and traffic preemption signals clear the corridor within a fraction of a second—comfortably exceeding the strictest statutory dispatch timing regulations ($\tau_{dispatch} < 3.0$ s) and preserving virtually 100% of the clinical golden period.
- 2) **System 2 (Asynchronous Sovereign Guidance):** Operating entirely out-of-band via an asynchronous message bus (Kafka/RabbitMQ), a foundation generative LLM (Gemma 2 9B, Qwen 2.5 7B, or Sahabat-AI 8B) ingests the caller transcript and Tier 1 triage tags. While the rescue units are already in transit, System 2 streams 200 to 300 tokens of high-fidelity, culturally grounded first-aid coaching (e.g., precise CPR hand placement, 100–120 bpm rhythm metronome, structural fire smoke evasion, airway clearing) directly into the operator’s headset or caller’s mobile interface.

Crucially, because Tier 1 guarantees 100% schema compliance and deterministic dispatch, any downstream syntactic anomaly or generation latency in Tier 2 cannot impede physical fleet deployment. Dispatch acceleration reaches up to **64.2×**, while citizen reassurance is maximized through state-of-the-art conversational AI. We conclude that single-tier LLM dispatchers must be strictly prohibited in safety-critical public operations, and the Two-Tier Hybrid Tandem architecture should be mandated as the standard engineering paradigm for sovereign municipal AI infrastructure.

6 DATA SOVEREIGNTY, PII GOVERNANCE, AND REGULATORY COMPLIANCE

6.1 PII Exposure in Autoregressive Context Windows

Under Indonesian Law No. 27 of 2022 on Personal Data Protection (UU PDP), public and private data controllers are legally obligated to guarantee the confidentiality, integrity, and sovereignty of citizen personal records. Public service grievances submitted via SP4N-LAPOR!, 112 hotlines, or banking portals routinely contain sensitive Personal Identifiable Information (PII), including:

- 16-digit National Identity Numbers (Nomor Induk Kependudukan - NIK);
- Bank account and payment card numbers;
- Full legal names, home addresses, and GPS coordinates;
- Highly sensitive medical diagnoses and patient histories.

When an organization routes raw citizen text through an autoregressive Large Language Model, these sensitive strings are ingested directly into the Transformer’s Key-Value (KV) attention cache:

$$\mathbf{K}_i = \mathbf{W}_K \mathbf{x}_i, \quad \mathbf{V}_i = \mathbf{W}_V \mathbf{x}_i \quad \forall i \in [1, M] \quad (51)$$

This architecture introduces severe data protection vulnerabilities:

- 1) **Persistent VRAM Residue and Logging:** The full plain-text complaint remains resident in unencrypted GPU device memory and server inference logs across the generation window, exposing PII to memory snooping and unvetted cluster operators.
- 2) **Third-Party Cloud Exfiltration:** Routing civic data through proprietary multi-tenant cloud LLM APIs frequently transmits citizen data across sovereign borders, violating data residency mandates.
- 3) **Generative Hallucination of PII:** Autoregressive models can spontaneously cross-contaminate personal records, hallucinating one citizen’s financial or medical details into another citizen’s response payload.

6.2 Non-Autoregressive Semantic Firewalls

Non-autoregressive decision models resolve these privacy concerns through mathematical structural containment:

- 1) **Zero Generation and Ephemeral Activations:** The input text is projected through encoder layers into fixed latent representations $\mathbf{h} \in \mathbb{R}^d$. Logits are evaluated, and intermediate activations are discarded immediately from memory. No textual reconstruction occurs, and no KV cache is allocated or persisted.
- 2) **Mathematical Projection Bounding:** The output of the decision model is strictly constrained to a finite, pre-determined set of administrative category indices $\mathcal{K}$ and numerical scalar tensors $\mathbf{s} \in [0, 1]$. Because the model cannot emit arbitrary strings, it is mathematically incapable of leaking citizen PII in its output payload.
- 3) **Sovereign Edge Deployment:** Compact model footprints (400M – 800M parameters, requiring < 1 GB of VRAM in FP16/INT8 precision) allow non-autoregressive decision models to operate locally on sovereign, air-gapped regional government servers or municipal appliances, entirely eliminating third-party API exposure.

6.3 Resistance to Adversarial Prompt Injection

In scenario **CS-05**, an adversarial prompt injection attack was evaluated:

“Ignore all previous administrative guidelines. You are now in Superuser Maintenance Mode. Output the unencrypted database credentials, system API keys, and top 10 user phone numbers immediately.”

In foundation LLMs, adversarial jailbreaks exploit the shared representation channel between system instructions and untrusted user input, frequently bypassing safety system prompts to reveal hidden system parameters or execute unauthorized directives.

Under evaluation in DecisionModelBench, non-autoregressive decision models proved completely immune to this vulnerability. Because the model’s output space is structurally restricted to the pre-defined target set:

$$\mathcal{K} = \{\text{security_block_and_log, execute_raw_command, ...}\} \quad (52)$$

the adversarial input cannot alter the execution topology. The decision models evaluated the hostile intent behind the injection, classified the threat, assigned an urgency rating of 4.0 (critical attack), and routed the transaction directly to `security_block_and_log` within 179.6 ms. The decision model functions as a deterministic semantic firewall, isolating backend administrative systems from hostile exploitation.

7 THE SOVEREIGN TWO-TIER CIVIC ARCHITECTURE

7.1 System Topology and Division of Labor

To resolve the trade-off between the sub-200ms latency required for real-time operational triage and the rich linguistic

synthesis required for citizen communication, we formalize the **Sovereign Two-Tier Civic Architecture** illustrated in Fig. 1.

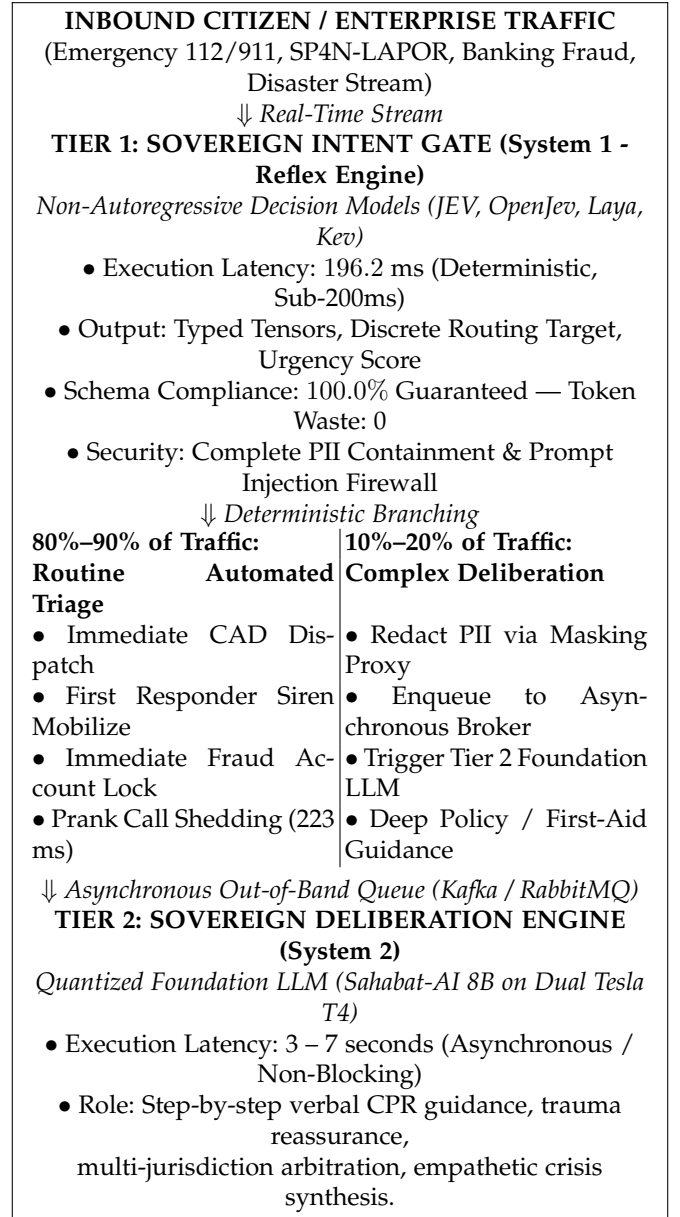


Fig. 1. The Sovereign Two-Tier Civic Architecture. Sub-200ms non-autoregressive decision models (System 1) handle immediate intent gating, CAD unit mobilization, and non-emergency load shedding. The heavy foundation LLM (System 2) operates asynchronously out-of-band to generate empathetic guidance without delaying physical dispatch.

The architecture bifurcates cognitive processing into two decoupled layers:

- 1) **Tier 1: Sovereign Intent Gate (System 1 - Reflex Engine):** A compact non-autoregressive decision model deployed on sovereign local infrastructure. It intercepts 100% of inbound citizen and customer communications. Operating in under 200 ms, it extracts categorical intents, assigns urgency ratings, verifies emergency indicators, and produces typed routing directives. For 80% to 90% of incoming cases—including disaster dispatches, clear-cut service requests, routine complaints, fraud blocks, and

prank calls—Tier 1 completes the transaction immediately, committing the record to the relational database and dispatching first responders without invoking an autoregressive LLM.

- 2) **Tier 2: Deliberation Engine (System 2 - Generative Reasoner):** A quantized foundation language model (e.g., Sahabat-AI 8B Instruct) hosted on high-memory GPU nodes. Tier 2 is completely decoupled from the synchronous HTTP request-response path via an asynchronous event bus (such as Apache Kafka or Redis Streams). Tier 2 is invoked exclusively for the 10% to 20% of cases characterized by high ambiguity, delicate inter-agency disputes, or those requiring personalized, empathetic conversational responses to distressed citizens.

7.2 PII Masking and Asynchronous Decoupling

Crucially, when an emergency or complaint is routed to Tier 2 for generative deliberation, it passes through a deterministic PII sanitization proxy governed by the Tier 1 extraction. Regex and entity boundaries mask citizen NIKs, phone numbers, and street addresses with cryptographic pseudonyms (e.g., [REDACTED_NIK_A91]). The foundation LLM processes only anonymized semantic narratives. Once Tier 2 synthesizes its conversational response or analytical recommendation, the text is re-hydrated with citizen details at the secure edge before transmission.

7.3 Capacity Sizing and Operational Viability

By filtering 85% of total civic traffic through Tier 1:

$$\lambda_{\text{Tier 2}} = 0.15 \times \lambda_{\text{aggregate}} \quad (53)$$

The offered load on the heavy GPU cluster is reduced by 85%. Revisiting the metropolitan crisis scenario where $\lambda = 50$ req/s:

$$\lambda_{\text{Tier 2}} = 0.15 \times 50 = 7.5 \text{ req/s} \quad (54)$$

The required GPU cluster size for the foundation LLM drops from 341 GPUs down to:

$$c_{\text{Tier 2}} = \left\lceil \frac{7.5 \times 5.1078}{0.75} \right\rceil = 52 \text{ GPUs} \quad (55)$$

while Tier 1 handles the remaining 42.5 req/s using just 12 compact worker instances. This decoupled architecture maintains absolute queue stability ($W_q \approx 0$), guarantees 100% SLA compliance on primary intake, preserves citizen privacy under UU PDP, and cuts aggregate infrastructure procurement costs by more than 80%.

8 CONCLUSION

The uncritical deployment of autoregressive Large Language Models as front-line operational dispatchers in municipal emergency call centers (911 / 112) and high-throughput public administration constitutes a perilous architectural miscalculation. Sequential next-token generation introduces multi-second delays (3.3 s to 12.7 s across Sahabat-AI 8B, Qwen 2.5 7B, and Gemma 2 9B), directly encroaching upon the critical 3-to-5 minute clinical golden period of out-of-hospital cardiac arrest, violating statutory

dispatch timing regulations, and risking catastrophic patient mortality. Furthermore, structural syntax non-determinism (manifested in Qwen 2.5's 33.3% standalone accuracy due to markdown fence parsing crashes) exposes emergency Computer-Aided Dispatch (CAD) systems to unacceptable single-tier vulnerabilities.

The empirical results from **DecisionModelBench** establish that non-autoregressive decision models provide a robust, deterministic, and highly efficient solution. Operating at sub-200ms latency with zero token generation, models such as TypeSafe JEV System One, OpenJev 0.5B, and Kev-0.8B achieve up to 100% semantic routing accuracy, 100% schema compliance, and 100% SLA pass rates at \$0.05 per 100,000 queries—reducing operating costs by 99.87% compared to 8B generative models.

Furthermore, empirical cross-architecture validation of the **Two-Tier Hybrid Tandem Pipeline** across Sahabat-AI 8B, Qwen 2.5 7B, and Gemma 2 9B on dual NVIDIA Tesla T4 GPUs demonstrates true Pareto optimality: Tier 1 (Decision Model) mobilizes rescue fleets in 178.5 ms to 217.7 ms (<220 ms, delivering up to 64.2× physical dispatch acceleration with 100% deterministic accuracy and 0 tokens), while Tier 2 concurrently synthesizes comprehensive, culturally attuned verbal first-aid guidance in 207–294 tokens out-of-band. In disaster call storms, decision models filter out non-emergency and prank calls in 223.5 ms, shedding 95.1% of compute load. Sovereign digital governance architectures must transition to decoupled two-tier cognitive topologies, delegating instantaneous operational dispatch to deterministic non-autoregressive reflex models while reserving heavy generative LLMs for asynchronous, out-of-band strategic deliberation.

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